

CSRD Compliance — Methodology & Audit Trail

Companion reference · reporting boundary, emission factors, data quality · IS6611 · Group 11

Reporting Boundary

Scope: Scope 2 (purchased electricity), location-based method only — no market-based instruments (PPAs/RECs) are in scope. **Facility:** single facility (ORNL Summit) — no multi-site consolidation. **Excluded:** Scope 1 (none applicable — no on-site combustion at a data centre) and Scope 3 (not modelled — no supply-chain or embodied-hardware data available). **Period:** the annual saving figure is a **projection** from a 5-day observed operational sample matched against 4 years of grid data, not a full-year direct measurement.

Savings Hierarchy — Reference Figures

Tier	Value (tCO2/yr)	Basis
Unconstrained potential	314 (sensitivity 209–418)	20–40% flexible-workload range around the 30% central assumption
Capacity-aware ceiling	87	Adds cluster node-capacity constraints (4,626 nodes)
Realistic, forecast-driven	34	Adds real 48h forecast error vs. perfect hindsight

Table 1 — The sensitivity range (209–418) applies to the unconstrained tier only; the capacity-aware and realistic tiers use the central 30% assumption throughout, as re-running the full capacity-aware/forecast-driven pipeline at each sensitivity level was out of scope.

Emission Factors Used

Fuel	Emission Factor (gCO2/kWh)	Category
Coal	1000	Fossil
Petroleum	800	Fossil
Natural Gas	450	Fossil
Other	500	Mixed
Nuclear	0	Zero-carbon
Hydro	0	Zero-carbon
Wind	0	Zero-carbon
Solar	0	Zero-carbon

Table 2 — IPCC AR5-style operational (generation-point) emission factors, applied to TVA's hourly generation-by-fuel mix to compute grid carbon intensity.

Data Quality & Provenance

ORNL Summit telemetry: ~99.98% complete, direct hardware measurement (NVIDIA DCGM for GPU power, Baseboard Management Controller for PSU draw) — not estimated. **TVA/EIA generation data:** ~99.6% complete, mandatory FERC Order 830 regulatory reporting — a legal obligation, not a voluntary estimate. Both sources were cross-validated against their own published documentation before use.

Methodology Summary

Carbon intensity is computed as the generation-weighted average of the emission factors above, applied hourly to TVA's actual fuel mix. Facility emissions are computed by pairing ORNL Summit's per-node

power draw (active/idle classified at a 250 W threshold) with the matching hour's grid carbon intensity. Savings are produced by a capacity-aware load-shifting optimiser (validated against a linear-programming benchmark) under a 30% flexible-workload central assumption, with 20–40% shown as a sensitivity range on the unconstrained tier above.

Carbon-Aware Scheduling for Data Centres · Group 11 · Companion to the CSRD Compliance dashboard view